

Dual Self-Injection Locking for Hybrid-Integrated Laser Linewidth Reduction and Stabilization

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Abstract Short on-chip external cavities improve laser stability but offer limited linewidth reduction, while long reflections narrow linewidth. We combine phase- and amplitude-controlled short and long cavities in a zero-process-change silicon-on-insulator circuit to simultaneously reduce linewidth and enhance stability against reflections in a hybrid-integrated DFB laser.

Introduction

Reflection-control circuits offer a practical route to semiconductor laser linewidth reduction and stabilization by shaping the amplitude and phase of controlled optical feedback [1–3]. Strong self-injection from a short external cavity has been shown to stabilize isolator-free DFB lasers against parasitic reflections [2], but it provides limited linewidth reduction compared to high- Q self-injection-locked [4] or extended-cavity [5] approaches. Here, we experimentally investigate the laser dynamics under controlled self-injection from short *and* long cavities in a silicon-on-insulator (SOI) circuit. We demonstrate that the short external cavity enables further linewidth reduction by the long cavity while achieving laser stability, reaching a minimum Lorentzian linewidth of 25 kHz, corresponding to a $32\times$ reduction compared to the free-running laser. In addition, we measure the sensitivity of the laser linewidth to the phase of self-injection within stabilized regimes. The entire circuit is implemented in a zero-process-change SOI platform, fully compatible with standard CMOS SOI silicon photonic fabrication, and requires no post-processing beyond hybrid integration with a laser. The design avoids the use of magneto-optic materials [6] and remains compatible with quantum well-based devices.

Device and Measurement Setup

The circuit schematic is depicted in Fig. 1. A DFB laser is hybrid integrated to the silicon chip via a photonic wire bond (PWB). Two waveguide Sagnac-loop mirrors form the short and long on-chip cavities. Each mirror is preceded by a Mach-Zehnder interferometer (MZI) that acts as a tunable power tap (TPT), controlling the amplitude of self-injection into each cavity independently. The phase of self-injection from the short and long

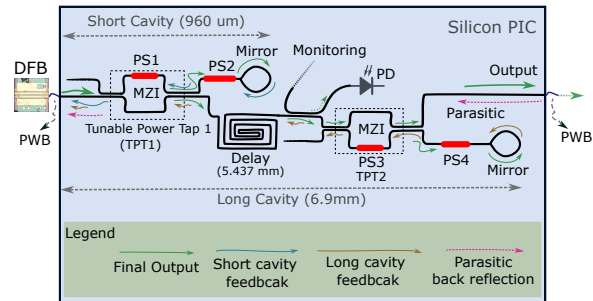


Fig. 1: Schematic of the dual-cavity reflection-control circuit. A DFB laser is hybrid integrated to the SOI chip via a photonic wire bond (PWB). Two tunable power taps (TPT₁, TPT₂) control the fraction of light diverted from the short and long Sagnac-loop mirrors, while phase shifters PS2 and PS4 set the round-trip phase of each cavity.

cavities, respectively, is controlled by thermo-optic phase shifters PS2 and PS4. We label each operating point by the TPT insertion losses (d_1, d_2) in dB, where d_1 is the power drop ratio of MZI₁ (short cavity) and d_2 is the power drop ratio of MZI₂ (long cavity), following the circuit shown in Fig. 1. A higher power drop value diverts more optical power from the corresponding cavity and less to the through port. Tunable power taps may not be required in a final implementation once suitable injection levels are established. In that case, only two phase shifters are needed to adjust the self-injection phase, enabling laser stabilization and optimal linewidth performance.

The total insertion loss of the two cascaded TPTs is $d_1 + d_2$; Table 1 lists the resulting maximum injection levels for each TPT operating bias point. The PWB interface introduces parasitic reflections estimated at -20 to -25 dB return loss at each edge coupler, which are typically sufficient to destabilize a DFB laser [2]. The short cavity provides a ~ 22 ps round-trip delay (including ~ 2.3 ps in the $230\ \mu\text{m}$ PWB, $n_g \approx 1.5$), while the long cavity has a round-trip delay of ~ 186 ps owing to its 5.4 mm on-chip delay spiral ($n_g \approx 4.0$). Both cavities share a

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Tab. 1: TPT operating bias points, injection levels at the laser facet, and measured best Lorentzian LW. d_1 and d_2 represent TPTs insertion losses incurred by the short- and long-cavity MZIs. Injection levels include all round-trip losses: MZI coupling, waveguide propagation (2.8 dB/cm), and the PWB (2×3 dB). The two reflections add in the dual-cavity mode; the net injection depends on the relative phase. The free-running LW without external feedback is ~ 800 kHz.

TPT bias (d_1, d_2)	Insertion loss (dB)	Short-cavity inj. (dB)	Long-cavity inj. (dB)	Best LW (kHz)
(1.5 dB, 0 dB)	1.5	-17.0	off	180
(0 dB, 1.5 dB)	1.5	off	-20.3	50
(1.0 dB, 2.0 dB)	3.0	-20.0	-20.3	25
(2.0 dB, 1.0 dB)	3.0	-14.9	-27.3	62

common bus waveguide, such that the laser sees a superposition of the two reflections, with their relative phase governed by the thermo-optic phase shifters PS2 (short cavity) and PS4 (long cavity). Notably, these time-delayed interactions give rise to dynamics that are more complex than those observed when each effect acts in isolation.

The measurement procedure for each operating point is as follows. First, all heaters are set to zero voltage, and the through-port PD power is recorded as a reference. MZI₁ is then tuned to the desired power drop d_1 using a calibration look-up table with closed-loop PD feedback. The reference power is re-read after biasing MZI₁, and MZI₂ is tuned to its target biasing point d_2 relative to this updated reference, ensuring that the two TPT settings are set sequentially and independently. With both MZIs biased, we sweep the phase shifters PS2 and PS4 on a 2D grid. At every grid point, we record an optical spectrum on the optical spectrum analyzer (OSA) and a delayed self-heterodyne (DSH) beat note (using 25 km fiber, 125 μ s delay) from which we extract the Lorentzian linewidth. The free-running DFB

linewidth without external feedback is ~ 800 kHz.

Results and Discussion

Table 1 summarizes the four operating points. TPT biasing points of (1.5 dB, 0 dB) and (0 dB, 1.5 dB) isolate the short and long cavities to characterize each feedback regime independently.

The short cavity (~ 22 ps round trip) places the laser well within the short-delay regime, where strong self-injection stabilizes the lasing mode and suppresses sensitivity to parasitic back-reflections, including the -20 to -25 dB PWB reflections at both edge couplers [2]. The long cavity (~ 186 ps round trip) accesses the narrow-linewidth [7], where the extended delay converts small frequency fluctuations into large phase excursions that are corrected by the feedback loop. The optical spectrograms of these two regimes as a function of the self-injection phase are shown in Fig. 2(a) and 2(d). Both long and short cavities can stabilize the laser into single-mode operation against small parasitic reflections from the edge couplers.

The best measured linewidth with only the short cavity engaged is 180 kHz, while the long cavity can reduce the laser linewidth to 50 kHz, despite

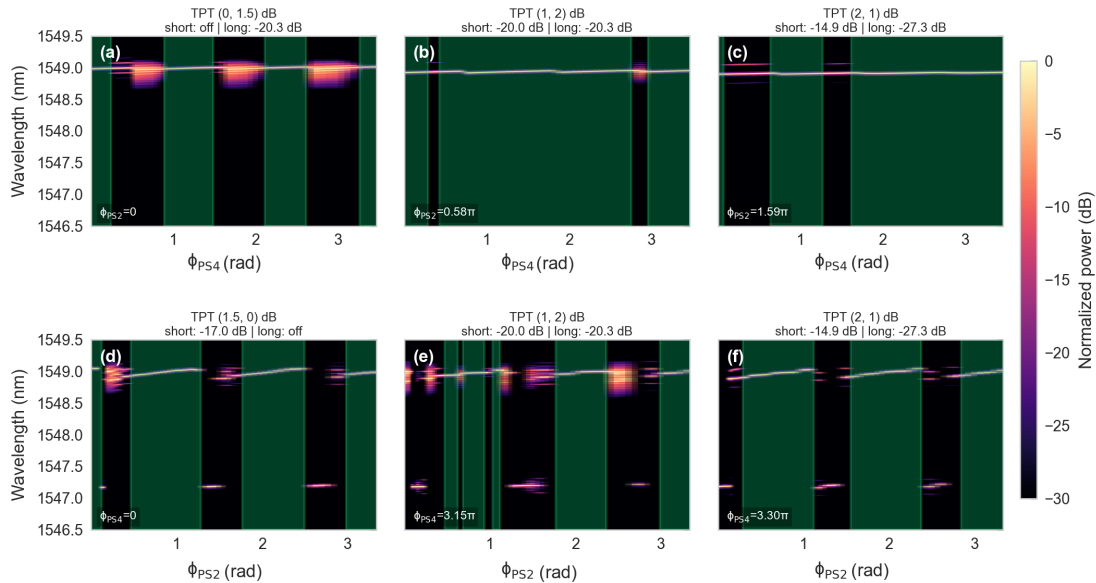


Fig. 2: Optical spectrograms of the four operating points listed in Table 1 as a function of self-injection phase (ϕ_{PS2} and ϕ_{PS4}). Each panel title reports the TPTs bias (d_1, d_2) and the resulting short- and long-cavity injection levels. The first column combines the two single-cavity cases: panel (a) shows the long-only point (0 dB, 1.5 dB) and panel (d) the short-only point (1.5 dB, 0 dB). Columns 2 and 3 show the dual-cavity points (1.0 dB, 2.0 dB) (panels (b),(e)) and (2.0 dB, 1.0 dB) (panels (c),(f)). Upper row (a–c): PS2 is fixed at its optimum value, and PS4 is swept. Lower row (d–f): PS4 is fixed at its optimum value, and PS2 is swept. The translucent green bands highlight phase ranges classified as single-mode operation at 1549 nm.

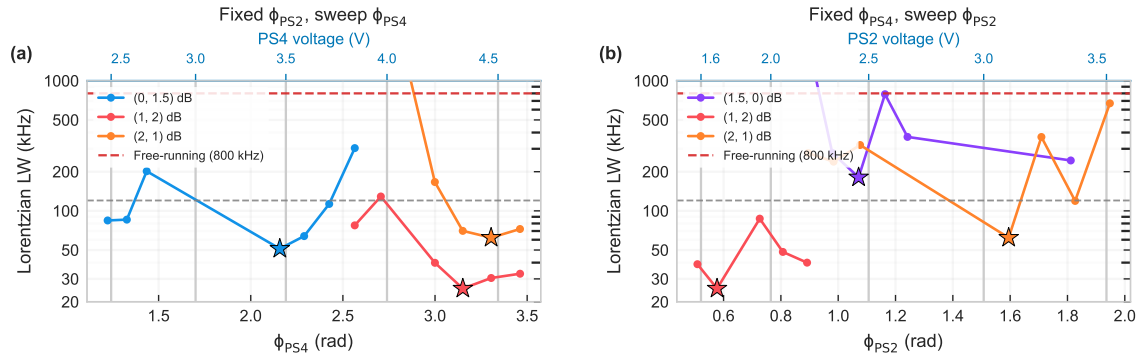


Fig. 3: Lorentzian linewidth as a function of self-injection phase (ϕ_{PS2} and ϕ_{PS4}) across TPTs biasing points. Panel (a): PS4 is swept with PS2 fixed at its optimum. Panel (b): PS2 is swept with PS4 fixed at its optimum. Star symbols mark the minimum linewidth in each sweep. The upper (blue) x-axis shows the actual heater drive voltage.

the additional 3 dB higher round-trip loss (additional 1.09 cm delay spiral at 2.8 dB/cm).

Without the stabilizing short cavity, the laser is also more susceptible to drifting into chaos during the phase sweep. This is visible in Fig. 2: the long-only case in panel (a) shows narrower green bands interspersed with unstable gaps as PS4 is swept, whereas the short-only case in panel (d) remains single-mode over a much wider phase range, as expected for short-delay self-injection locking. The instability of the laser in the long-cavity regime is characterized by a significantly broadened spectrum, whereas the short-cavity regime leads to multi-mode and pulsation operation.

Dual-cavity operation combines the stabilizing effect of the short cavity with the linewidth reduction capabilities of the long cavity. At TPT's biasing point of (1.0 dB, 2.0 dB), where both cavities inject nearly equal power (−20.0 dB and −20.3 dB), the linewidth reaches 25 kHz—the best measured value—with an extensive single-mode coverage in panels (b) and (e) of Fig. 2 and a broad low-linewidth region over PS4 in Fig. 3(a). At TPT's biasing point of (2.0 dB, 1.0 dB), the short cavity provides stronger injection (−14.9 dB) that anchors the laser in a stable single-mode state, while a weaker long-cavity contribution (−27.3 dB) narrows the linewidth to 62 kHz. By comparing these two biasing points, it can be seen that the longer cavity contributes more to the linewidth reduction, as expected.

Fig. 3 shows the laser linewidth across stabilized regions as a function of the self-injection phase. In Fig. 3(a), the best achievable linewidth occurs when the reflection from the long cavity is strongest (TPTs biased at (1.0 dB, 2.0 dB)), with linewidths below 100 kHz maintained across the stable region. Fig. 3(b) presents the linewidth as a function of the short-cavity self-injection phase. At the same TPT bias point (1.0 dB, 2.0 dB), a sub-100 kHz linewidth is maintained over the full measured PS2 range. The near-continuous stable band in Fig. 2(b) further confirms that the short cavity establishes a robust operating basin, within

which the long cavity can be tuned to achieve additional linewidth narrowing.

These results demonstrate that the TPT biasing provides continuous control over the trade-off between stability (primarily governed by short-cavity injection) and linewidth reduction (primarily governed by long-cavity delay), enabling precise shaping of the laser dynamics without requiring modifications to the photonic circuit.

Conclusions

We experimentally demonstrated that combining short- and long-delay self-injection in a dual-cavity SOI reflection-control circuit enables simultaneous laser stabilization and linewidth narrowing in the presence of parasitic reflections, which are inherent to any PIC assembly. The short cavity provides robust single-mode stability, while the long cavity enables deeper linewidth reduction. The best Lorentzian linewidth of 25 kHz was achieved when both cavities were active, a $3.4\times$ improvement over the short cavity alone, $2.0\times$ over the long cavity alone, and $32\times$ over the free-running DFB. The two TPT bias points offer continuous, independent control of injection strength into each cavity, allowing the trade-off between stability and linewidth to be tuned electrically on a single photonic chip. The entire circuit is fabricated in a standard CMOS-compatible SOI process, making it readily scalable. Future work will focus on extending the long cavity by $10\times$ to $100\times$ to investigate the achievable linewidth reduction with increased feedback delay. In parallel, we aim to develop a reconfigurable chip incorporating multiple cavities (e.g., 2–4) with logarithmically increasing lengths, enabling experiments on different types of laser coherence and providing a platform for further laser dynamics studies.

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